SIMATS ENGINEERING

CO5 - ASSESSMENT TOOL 2

Mock Interview

COURSE NAME: Deep Learning **COURSE CODE: MLA0408**

COURSE OUTCOME COVERED

CO5: Students will be able to evaluate advanced machine learning concepts such as reinforcement learning, Monte Carlo methods, sampling techniques, policy iteration, value iteration, and temporal difference learning. They will understand computational learning theory concepts including VC dimension, sample complexity, and mistake-bound analysis. Students will be capable of selecting and optimizing advanced learning strategies for intelligent systems and complex applications. (BL-6)

Assessment Tool Weightage: 20%

QUESTIONS:

Q1. Autoencoder Design

- What is an autoencoder and what are its main components in a neural network architecture?
- How would you design an autoencoder for reducing the dimensionality of a high-dimensional dataset?
- What factors would you consider when deciding the size of the encoder and decoder layers?
- Why is the bottleneck layer important in an autoencoder design?
- How would you choose an appropriate activation function for the encoder and decoder of an autoencoder?

Q2. Stochastic Encoders and Decoders

- What is a stochastic encoder, and how does it differ from a deterministic encoder?
- Why is randomness introduced into the encoding process of a stochastic autoencoder?
- How does a stochastic decoder reconstruct data from a probabilistic latent representation?
- What are the advantages of using stochastic encoders and decoders in generative models?
- How does a stochastic encoder represent uncertainty in the input data?

Q3. Deep Belief Network vs Deep Boltzmann Machine

- What is the main architectural difference between a Deep Belief Network (DBN) and a Deep Boltzmann Machine (DBM)?
- How are connections between hidden layers organized differently in a DBN and a DBM?
- Why is a DBN considered a hybrid model containing both directed and undirected connections?
- How does a DBM differ from a DBN in terms of the direction of connections between layers?
- How does the training procedure of a DBN differ from that of a DBM?

Q4. Generative Adversarial Networks

- What is a Generative Adversarial Network, and what are its two main components?
- How does the generator in a GAN create new data samples?

- What is the primary role of the discriminator in a GAN?
- How do the generator and discriminator compete with each other during GAN training?
- Why is balancing the generator and discriminator important for stable GAN training?

Q5. Integrated Deep Learning Solution

- What do you understand by an integrated deep learning solution, and what are its major components?
- How would you combine different deep learning models to solve a complex real-world problem?
- How would you decide whether to use an autoencoder, DBN, DBM, or GAN in an integrated solution?
- What factors should be considered when designing an end-to-end deep learning pipeline?
- How can feature extraction and representation learning be integrated into a deep learning solution?

Code of Conduct:

I T. NarasimhaReddy(192425190)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: T. Narasimha Reddy

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 Exemplary	Level 3 Proficient	Level 2 Developing	Level 1 Beginning
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving	25	Solves problems logically and applies concepts effectively in real scenarios	Applies concepts with moderate accuracy	Limited problem-solving ability; requires guidance	Unable to solve or apply concepts
Analytical Thinking	15	Analyzes questions critically and provides well-structured, logical answers	Provides reasonable analysis with some structure	Superficial or partially structured responses	No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence	15	Displays high confidence, positive attitude, and professional behavior throughout	Shows good confidence with minor issues	Low confidence; inconsistent behavior	Lacks confidence and professionalism

SIMATS ENGINEERING

CO5 - ASSESSMENT TOOL 2 (Mock Interview)

Course: Deep Learning (MLA0408)

Q1. Autoencoder Design

What is an autoencoder and its main components?

An autoencoder is an unsupervised neural network that learns to reconstruct its own input. Its main components are an encoder (compresses input to a latent code), a bottleneck/latent layer (compact representation), and a decoder (reconstructs the input from the latent code).

How would you design one for dimensionality reduction?

Use an undercomplete architecture where the bottleneck has far fewer units than the input, train it with reconstruction loss (MSE) on the data, then use the bottleneck activations as the reduced-dimension features for downstream tasks.

Factors for sizing encoder/decoder layers

Input dimensionality, amount of training data, desired compression ratio, computational budget, and the complexity of patterns in the data all guide layer sizes; layers are usually reduced gradually rather than in one large jump.

Why is the bottleneck layer important?

It forces the network to discard redundant information and keep only the most informative features, preventing the autoencoder from simply learning an identity mapping and enabling meaningful compression.

Choosing activation functions

ReLU is typically used in hidden encoder/decoder layers for efficient gradient flow, while the output layer uses sigmoid (data scaled to [0,1]) or linear activation (unbounded real-valued data) to match the input's range.

Q2. Stochastic Encoders and Decoders

What is a stochastic encoder and how does it differ from a deterministic one?

A stochastic encoder outputs the parameters of a probability distribution (e.g. mean and variance) and samples the latent code from it, whereas a deterministic encoder maps each input to a single fixed latent vector.

Why introduce randomness into encoding?

Randomness prevents the model from overfitting to exact input values, encourages smoother and more generalisable latent spaces, and allows the model to represent uncertainty in the data.

How does a stochastic decoder reconstruct data?

It takes a sampled latent code and outputs the parameters of a distribution over the output space, then reconstructs data by sampling from (or taking the expected value of) that distribution rather than producing a single fixed output.

Advantages in generative models

They enable smooth, continuous latent spaces that support sampling new data, better generalisation, robustness to noise, and principled uncertainty estimation, which is essential for realistic data generation.

How does it represent uncertainty?

The variance (spread) of the output distribution reflects the model's confidence: a narrow distribution indicates high certainty about the input, while a wide distribution indicates ambiguity or noise in the input.

Q3. Deep Belief Network vs Deep Boltzmann Machine

Main architectural difference

A DBN has directed connections between most layers with only the top two layers undirected, while a DBM has undirected connections between every pair of adjacent layers throughout the network.

How are hidden-layer connections organised differently?

In a DBN, lower layers form a directed sigmoid belief network fed by the top undirected RBM; in a DBM, all layers are symmetrically connected undirected RBMs, allowing top-down and bottom-up influence between every layer pair.

Why is a DBN a hybrid model?

Because its top two layers form an undirected RBM while the remaining lower layers use directed, generative (belief network) connections, combining both connection types in one architecture.

Direction of connections: DBM vs DBN

A DBM uses purely undirected connections across all layers, whereas a DBN uses directed connections for all but its topmost layer pair.

Difference in training procedure

A DBN is trained via greedy layer-wise pre-training of stacked RBMs followed by optional fine-tuning; a DBM is trained jointly across all layers using approximate inference (e.g. mean-field) combined with contrastive-divergence-style updates, which is more computationally demanding.

Q4. Generative Adversarial Networks

What is a GAN and its two main components?

A GAN is a generative model made of two competing networks: a generator, which creates synthetic data from random noise, and a discriminator, which tries to distinguish real data from generated data.

How does the generator create new samples?

It takes a random noise vector as input and passes it through upsampling/transposed-convolution layers to transform the noise into a structured, data-like output such as an image.

Primary role of the discriminator

It acts as a binary classifier that estimates whether a given sample is real (from the training data) or fake (produced by the generator), providing the training signal that pushes the generator to improve.

How do generator and discriminator compete?

They play a minimax adversarial game: the discriminator is trained to better detect fakes while the generator is trained to fool the discriminator, with each network's improvement pushing the other to improve in turn.

Why is balancing them important?

If the discriminator becomes too strong too quickly, the generator receives weak or vanishing gradients and stops improving; if the generator dominates, the discriminator cannot provide a useful learning signal, so both must improve at a comparable pace for stable training.

Q5. Integrated Deep Learning Solution

What is an integrated deep learning solution and its major components?

It is a pipeline that combines multiple deep learning models/techniques to solve a complex problem end-to-end, typically including data pre-processing, feature extraction/representation learning, one or more core generative or discriminative models, and an output/decision layer.

How would you combine different models for a complex problem?

Use each model for the sub-task it is best suited to, for example an autoencoder or DBN for unsupervised feature learning, followed by a supervised or generative model (e.g. GAN) built on top of those learned features, chaining their outputs together in a single pipeline.

How to decide between autoencoder, DBN, DBM, or GAN?

The choice depends on the goal: autoencoders for compression/denoising/feature extraction, DBNs/DBMs for learning hierarchical generative representations from largely unlabelled data, and GANs when the priority is generating new, realistic samples rather than reconstructing existing ones.

Factors in designing an end-to-end pipeline

Data quality and volume, computational resources, latency/scalability requirements, interpretability needs, and how well each candidate model's strengths match the specific sub-task all influence the pipeline design.

Integrating feature extraction and representation learning

Unsupervised models (autoencoders, DBNs) can be pre-trained on raw data to learn compact, informative representations, and these learned embeddings are then fed as input features into the next stage of the pipeline, improving the efficiency and accuracy of the overall solution.